# Software Quality Assurance

Long Questions · Short and easy answers · From the lecture slides

| # | Long question | Priority |
|---|---|---|
| 1 | QA vs QC | Most important |
| 2 | Cost of Quality | Most important |
| 3 | Benchmarking Process | Backup |
| 4 | Garvin's Quality Perspectives | Backup |
| 5 | Dimensions of Quality | Backup |
| 6 | McCall Quality Model | Backup |
| 7 | Customer Satisfaction and Quality | Read once |

**How to write every long:** Definition → Table → Conclusion. Draw the table neatly, underline keywords, and add an example.

## LONG 1: QA vs QC

PPT: SQA vs SQC · slides 6–20

### Definition

**Quality Assurance (QA):** QA is **process oriented** and focuses on **preventing defects**. It ensures quality in the **process** used to make the product.

**Quality Control (QC):** QC is **product oriented** and focuses on **finding defects**. It ensures quality in the **product** by finding and removing defects.

| Basis | QA | QC |
|---|---|---|
| **Oriented to** | Process | Product |
| **Approach** | Proactive | Reactive |
| **Focus** | Prevent defects | Find defects |
| **Responsibility** | Everyone in the team | Testing team |
| **Examples** | Audits, standards, checklists, training | Testing, debugging, inspection, walkthrough |

**Conclusion:** QA **prevents** defects and QC **finds** them. Both are needed for a quality product.

**Yaad karne ki trick: A**ssurance = **A**age se bachao (Process, Prevent, Proactive)
**C**ontrol = **C**hecking kar ke pakro (Product, Find, Reactive)

**8 marks ke liye yeh line bhi likho: ISO:** QA provides **confidence** that quality requirements will be fulfilled. QC focuses on **fulfilling** quality requirements.

## LONG 2: Cost of Quality

PPT: Cost of Quality · slides 2–8

### Definition

Cost of Quality (COQ) is the **cost of defects**. It is the money that would **not be spent** if the product were made perfectly.

| Category | Meaning | Examples |
|---|---|---|
| **1. Prevention** | Cost to **stop** defects | Training, process improvement |
| **2. Appraisal** | Cost to **find** defects | Testing, audits |
| **3. Internal Failure** | Fixing defects **before delivery** | Rework, regression testing |
| **4. External Failure** | Fixing defects **after delivery** | Warranty, returned product, complaints |

**Important:**

• Coding, installation and shipping are **not** COQ.

• Spending more on **prevention and appraisal** reduces failure costs.

**Conclusion:** Preventing defects early saves money, because fixing them after delivery is very costly.

**Yaad karne ki trick: P-A-I-E**: **P**ehle roko, **A**ankho, **I**ndar theek karo, **E**xternal theek karo

**8 marks ke liye yeh line bhi likho:** External failure cost is the **most expensive**, because defects cost much more to fix after delivery.

## LONG 3: Benchmarking Process

PPT: Quality · slides 27–37

### Definition

Benchmarking is a process in which an organization **learns the best practices of other organizations** and adopts them to **improve** its own processes, products and services. It helps the organization **"think outside the box"**.

| Step | What happens |
|---|---|
| **1. Determine what to benchmark** | Choose the area to improve (process, project, product or service) |
| **2. Establish infrastructure** | Get a **sponsor** and form a **benchmarking team** |
| **3. Understand current practices** | Study your own current work and key performance factors |
| **4. Find source of information** | Find the **best practice leader** |
| **5. Analyze best practices** | Compare with the leader and find **performance gaps** |
| **6. Plan and implement improvement** | Make a plan, test it (pilot) and apply it |

**Conclusion:** Benchmarking is a **continuous process**. The steps are repeated to keep improving.

**Yaad karne ki trick: Kya → Team → Apna → Kis ka → Muqabla → Sudhaar**

## LONG 4: Garvin's Quality Perspectives

PPT: Quality · slides 11–16

### Definition

Garvin described **five perspectives** (views) of quality.

| Perspective | Meaning | Example / By |
|---|---|---|
| **1. Transcendental** | Quality can be **recognized but not defined**: "I know it when I see it" | Tom Peters |
| **2. Manufacturing** | Quality = **conformance to specification** | Philip Crosby |
| **3. User** | Quality = **fitness for use** | Joseph Juran |
| **4. Product** | Quality depends on the product's **characteristics ("ilities")** | Reliability, usability |
| **5. Value-based** | Quality depends on **how much the customer will pay** | Video game vs. airplane software |

**Conclusion:** Quality has no single definition. It depends on **who** is looking at it.

**Yaad karne ki trick: T-M-U-P-V**: "**T**um **M**ere **U**stad **P**e **V**ishwas karo"

## LONG 5: Dimensions of Quality

PPT: Quality · slides 6–9

### Definition

Dimensions of quality are the **8 features** customers use to judge the quality of a product.

| Dimension | Question |
|---|---|
| **1. Performance** | Does it do the job? |
| **2. Reliability** | How often does it fail? |
| **3. Durability** | How long does it last? |
| **4. Serviceability** | How easy is it to repair? |
| **5. Aesthetics** | How does it look? |
| **6. Features** | What extra does it do? |
| **7. Perceived Quality** | What is the company's reputation? |
| **8. Conformance to Standards** | Is it made exactly as designed? |

**Conclusion:** A product is of high quality when it is good in these dimensions.

**Yaad karne ki trick:** "**P**akistan **R**ail **D**er **S**e **A**ati, **F**ir **P**hir **C**hal parti" (P-R-D-S-A-F-P-C)

## LONG 6: McCall Quality Model

PPT: Software Quality Attributes · slides 16–20

### Definition

A quality model lists the characteristics that good software must have. The **McCall model** groups quality factors into **three categories** based on how software is used.

| Category | Meaning | Factors |
|---|---|---|
| **1. Product Operation** | While **using** the software | Correctness, Reliability, Efficiency, Integrity, Usability |
| **2. Product Revision** | While **changing** the software | Maintainability, Flexibility, Testability |
| **3. Product Transition** | While **moving** it to a new environment | Portability, Reusability, Interoperability |

**ISO 9126** (another model) has **6 characteristics**: Functionality, Reliability, Usability, Efficiency, Maintainability, Portability.

**Conclusion:** Quality models help define and measure software quality.

**Yaad karne ki trick:** Operation = **chalana**, Revision = **badalna**, Transition = **le jaana**
ISO 9126 = **"FRUEMP"**

## LONG 7: Customer Satisfaction and Quality

PPT: Quality · slides 23–26

### Definition

Customer satisfaction depends on **three levels** of quality.

| Level | Meaning | Example (Car) |
|---|---|---|
| **1. Basic Quality** | Customer **assumes** it is there and doesn't ask. Missing it causes **dissatisfaction** | Tyres, steering wheel |
| **2. Expected Quality** | Customer **asks** for it. More quality means more **satisfaction** | Model, colour |
| **3. Exciting Quality** | **Unexpected** features the customer didn't know they wanted. Always **satisfies** | Cup holder (when new) |

**Conclusion:** "**Today's innovations are tomorrow's expectations.**" Cup holders were once exciting, now they are basic.

**Yaad karne ki trick:** **B**asic = **B**ina kahe chahiye, **E**xpected = **E**xplicitly maanga, **E**xciting = **E**k surprise